

JEE MAIN REVISION MASTER: CENTER OF MASS AND COLLISION 2026 0 QUESTIONS 0% WEIGHTAGE ↓100%

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 8 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Center Of Mass - Core Application	8 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Center Of Mass - Core Application (High Yield — 8 PYQs)

Core Principle: Key problem pattern in Kcet Center Of Mass.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "During an elastic collision between two bodies, which of the following statements are correct? I. The initial kinetic energy is equal to the..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Three particles of mass 1kg,2kg and 3 kg are placed at the vertices A, B and C respectively of an equilateral triangle ABC of side 1 m . T..." [Ans: N/A]
- [2022 Shift] Q#3 (MCQ): "The centre of mass of an extended body on the surface of the earth and its centre of gravity..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2025 (Q#1)

During an elastic collision between two bodies, which of the following statements are correct? I. The initial kinetic energy is equal to the final kinetic energy of the system. II. The linear momentum is conserved. III. The kinetic energy during Δt (the collision time) is not conserved.

- (A) II and III only
- (B) I and III only
- (C) I, II and III
- (D) I and II only

Correct Answer: None

Solution Analysis: I. Final total kinetic energy equals initial II. Linear momentum is always conserved III. During collision kinetic energy gets partly converted to potential energy $\therefore$ All statements are correct